An open-source, provenance-aware framework for reconstructing missing CT dose indices from DICOM metadata

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Abstract

Background: The recorded volume CT dose index (CTDIvol) is often absent from de-identified image headers, and public archives carry no radiation dose structured report (RDSR), leaving retrospective dose work no recorded value to use.

Purpose: To present an open-source, provenance-aware framework that recovers dose indices from the metadata surviving de-identification, keeps a recovered value distinguishable from a recorded one, and measures the limits of that recovery.

Methods: An open-source Python package (MIT License) computes CTDIvol, dose-length product (DLP), size-specific dose estimate and effective dose. Every value carries its formula, that formula's inputs, the assumptions they rested on, and a label stating whether it was read from the data or rebuilt from it. Where no recorded index survives, CTDIvol is rebuilt from tube current, rotation time, collimation and pitch using an openly licensed table of normalized CTDI values. I parsed 14 RDSR objects, screened 400 public CT series, and compared the reconstruction against two references — 46

archive series retaining a CTDIvol, and 5 GE irradiation events recording one — both console figures, and so not independent of that table.

Results: All 14 RDSR objects parsed, and both consistency checks held in 14 of 14 files. CTDIvol was absent from the image header in 206 of 400 series (51.5%), ranging from 24.0% to 100.0% by manufacturer; of those 206, 42 (20.4%) could still be reconstructed, making 236 of 400 (59.0%) dose-recoverable. Against the archive's own recorded values (n = 46) the median relative error was −8.7%, and −5.7% (95% CI −12.4% to +7.8%) where the header recorded the total collimation; 18 of the 46 were over-estimates and 28 under-estimates, so the error is two-sided rather than conservative. On GE, which supplies no archive reference, five events from the scanners' own dose reports were compared: their exposure time refutes the collimation four of them record, leaving that width undetermined and the error spanning −18.2% to +22.4%; the fifth, whose recorded geometry it corroborates, spans −0.0% to +12.7% over the unrecorded bowtie filter alone.

Conclusion: Dose indices can be recovered when no recorded value survives, but only as far as the metadata and the open coefficient table allow, and provenance keeps a recovered value from being read as a recorded one.

1. Introduction

Computed tomography is the largest single contributor to collective medical exposure in most surveyed countries [1], and optimization rests on CTDIvol and the dose-length product (DLP), phantom-referenced indices of scanner output [2]. Modern scanners record them in a DICOM Radiation Dose Structured Report (RDSR) [3], which dose-management systems such as OpenREM harvest from clinical PACS [4]. That infrastructure does not transfer to public imaging archives: de-identified collections generally carry no RDSR and often omit CTDIvol from the header too, so a study relating image findings to the doses that produced them has no recorded value to work from. AAPM Report 246 catalogues the DICOM attributes organ dose estimation requires [5], but how often they survive de-identification is, to my knowledge, unreported.

Where the recorded index is missing, the surviving acquisition description suffices to reconstruct

CTDIvol from a scanner's normalized CTDI coefficients, as dose-evaluation software has done since CT-Expo [6]. Such a reconstruction is only as accurate as its metadata and its coefficients, so its uncertainty must be measured rather than assumed, and both code and coefficients must be open. This note describes a framework meeting those conditions and measures how far it reaches.

2. Materials and methods

2.1 Dose engine and the provenance labels

`ctdose-core` has a functional API and a command-line interface, and depends only on `pydicom`, `numpy` and `scipy`. Every quantity it returns carries four levels of provenance: value and unit; the formula; that formula's inputs and any assumption they rested on; and a source label, used throughout with these meanings:

- **recorded** — read from the data as the scanner wrote it, from an RDSR or from the CTDIvol attribute (0018,9345);
- **reconstructed** — absent from the data and rebuilt from the acquisition description and a published coefficient table (§2.3);
- **derived** — computed from a quantity already obtained by either route, such as a DLP from a CTDIvol;
- **unrecoverable** — neither present nor rebuildable, a required attribute being absent or the scanner untabulated. An outcome, not an error.

A series can offer a CTDIvol in more than one place, and the places do not always agree. The candidates are ranked by how directly each records what the scanner delivered — RDSR, image header (0018,9345), dose comment (0040,0310), reconstruction — and the highest is used. **The lower-ranked candidates are not discarded**: each is kept with its numerical difference from the chosen value, so a disagreement between two records of one exposure is visible rather than resolved silently.

2.2 RDSR parsing and its consistency checks

The parser is written against the coded content items of DICOM PS3.16 templates TID 10011–10013 [3], not any vendor's layout, extracting the per-event CTDIvol, DLP, reference phantom and technique

plus the study-level accumulated DLP and event count. Because an RDSR describes an examination rather than one series, attributing an event to a series is explicit — protocol name first, scanning length second — and where no event matches unambiguously, nothing is taken.

Two checks are computed for every object rather than assumed: the per-event DLPs are summed and compared with the accumulated DLP the report records separately, read by an independent traversal, and agreement within 1% is required; and the number of events parsed is compared with the total the report declares. Both reference quantities come from different parts of the object from the ones they check, so agreement is evidence neither was misread.

2.3 Physics reconstruction when no recorded dose exists

Where no recorded CTDIvol is available, it is reconstructed from an openly licensed table of normalized CTDI values: Dinwiddie et al. published under CC BY 4.0 the weighted CTDI per 100 mAs at pitch 1 for 17 scanner models from four manufacturers, tabulated by tube voltage, total collimation, reference phantom and bowtie filter [7]. Those values are what the scanner consoles reported, and a console figure is pre-set by the manufacturer from in-factory measurements [7] — a survey of what these scanners display, not an independent measurement of them, which bounds what §2.7 and §4 can claim. From one row of that table,

$$\mathrm{CTDI_{vol}} = \frac{n\mathrm{CTDI_w}}{100} \cdot \frac{I t_{rot}}{p}, \quad \mathrm{DLP} = \mathrm{CTDI_{vol}} \cdot \frac{L_{img}}{10}, \quad E = \mathrm{DLP} \cdot k,$$

with $n\mathrm{CTDI_w}$ the tabulated value for the series' voltage, collimation, phantom and bowtie, I the mean tube current, t_{rot} the rotation time, p the pitch, L_{img} the imaged scan length in mm, and k the region- and age-specific DLP-to-effective-dose coefficient of AAPM Report 96, Table 3 [8]. Those coefficients rest on ICRP Publication 60 tissue weighting, by way of the European Guidelines and the Monte Carlo organ doses Report 96 derived them from; ICRP Publication 103 [9] revised several weighting factors, so an effective dose reported here is not an ICRP 103 quantity and says so in its own provenance. The sex-specific ICRP 103 coefficients of Deak et al. [10] are also shipped. Because the table is tabulated at the voltage and collimation in question, neither an over-beaming factor nor a voltage extrapolation enters.

The reconstructed DLP uses the imaged scan length and so omits helical over-ranging, the extra

rotations at each end that irradiate beyond the reconstructed volume. For a given CTDIvol this biases the DLP low, but does not make it an underestimate overall, because the CTDIvol it is built on carries a two-sided error of its own (§3.3), so neither a DLP nor an effective dose from one is a bound in either direction. The omission is recorded as an assumption on the value rather than presented as a direction of error.

Selecting a row from the coefficient table of Dinwiddie et al. [7] is explicit and refusable. Manufacturer and model must match a tabulated model exactly or through aliases denoting the same machine; a family name identifying no single model is reported as unknown, not costed against a nearby one. A tube voltage between two tabulated values is interpolated as a power law and **refused** outside the tabulated range. An untabulated collimation takes the nearest one. No DICOM attribute records the bowtie filter, so the one conventional for the phantom is assumed — small at 16 cm, medium at 32 cm. Every such substitution is recorded on the value.

2.4 Size-specific dose estimate

The water-equivalent diameter is computed from an automatically extracted body contour as $D_w = 2\sqrt{(\overline{\mathrm{HU}}/1000 + 1)A_{ROI}/\pi}$ [11], using a fixed attenuation threshold, largest-connected-component selection and hole filling, so it is deterministic and needs no deep-learning dependency; SSDE is $\mathrm{CTDI}_{\mathrm{vol}} \cdot f(D_w)$, f from the AAPM TG-204 tables [12].

2.5 Archive sample, and what makes a series recoverable

The archive sample was drawn from The Cancer Imaging Archive [13] over its public NBIA interface, selected by DICOM `Manufacturer` and sampled round-robin: 400 CT series, 100 per manufacturer, across 92 collections.

A series is counted as **dose-recoverable** if a dose index can be obtained by either route: it carries a recorded CTDIvol, or no recorded value survives but the reconstruction of §2.3 can run on it. The latter needs two independent conditions at once — the acquisition described completely enough, meaning tube current, rotation or exposure time, total collimation, slice thickness and slice locations all present, **and** the scanner model present in the coefficient table. A series meeting neither is unrecoverable. These

conditions are read from the header alone and involve no dose calculation, so they are reported separately from accuracy.

2.6 Accuracy against the archive's own recorded values

A series already carrying a recorded CTDIvol supplies a reference the archive provides itself: the reconstruction is run as though the value were missing and compared with it. Of the 194 screened series carrying one, 53 sat on a scanner in the coefficient table, and 46 of those also had complete acquisition parameters. **All 46 were evaluated**: the eligible set is defined by the table's coverage rather than sampled, so there is no cap and no seed. Implausible recorded values were excluded beforehand by a physical-range test, not by eye. The pre-specified primary stratum is the subset whose header records the total collimation, a substituted width confounding the dose model with the archive's metadata gaps. Relative error is (reconstructed − recorded) / recorded, so the reference stands in the denominator, and confidence intervals are percentile bootstrap intervals over cases.

2.7 Accuracy on GE, against the scanners' own dose reports

No GE series in the archive sample carries a recorded CTDIvol (§3.2), so a second reference was taken from the RDSR objects of §2.2, which record both the CTDIvol a GE scanner computed and the technique behind it. Eligibility was fixed before any prediction and every excluded event is counted with its reason: an event enters if it records a CTDIvol, if its scanner is tabulated, if the table moves — spiral and sequenced both qualify, stationary does not — and if the technique is complete. Supplementary §S2.5 gives the criteria in full with the exclusions.

The report carries a measurement that constrains the geometry it claims. DCM 113824 records the event's total exposure time T, independent of the collimation the report states, and DICOM defines the scanning length L (113825) as the table travel over the entire tube loading, so the two span the same interval and the table speed gives $w = Lt_{rot}/(Tp)$ — the width the exposure used, from four recorded numbers. Nothing is corrected: w is reported beside the recorded one.

w refutes, and does not select. The four refuted events differ from their measured width by 86% to 87% and the corroborated one by 0.6%, so saying the recorded field is not a beam width needs no precise w; saying which tabulated row the scan used would, and nothing in the report establishes that

(§S2.5). So where W contradicts the recorded width the collimation is undetermined and the span runs over every tabulated row, with the narrower span that follows from applying the same margin to the candidate rows reported beside it, since the report does not determine which reading to take. Where W agrees, the recorded width stands. **DICOM records no bowtie filter**, so it varies throughout.

What this arm does and does not establish. It tests the dose model — table coefficient, mAs per rotation, pitch — and does **not** test recovery from a de-identified header, the technique here coming from a structured report rather than a stripped one. Nor does it test the vendor's dosimetry, the reference and the coefficients both being console figures from the same factory calibrations (§2.3). The events are not independent patients: several share a study, scanner and session.

2.8 Data and reproducibility

The RDSR set is the sample collection distributed with OpenREM (GPL-3.0) [4]: 15 objects from four manufacturers, 14 of them CT RDSRs, read from a user-supplied directory, not redistributed.

Every number reported in this note is written into a result file by the script that computes it, and those files ship with the software. The manuscript's values are checked against them by the automated test suite on every change, so a value edited by hand or left by an earlier run disagrees with its file and fails. The figures are regenerated from the same files.

3. Results

3.1 Recorded dose is read correctly across manufacturers

Of 15 sample objects, 14 parsed as CT Radiation Dose Structured Reports; the fifteenth is a secondary-capture dose screen and was correctly rejected. The 14 span 4 manufacturers (GE, Siemens, Philips, Canon/Toshiba) and 9 scanner models, holding 67 irradiation events of which 44 carry a CTDIvol spanning 0.14–222.59 mGy; both reference phantoms occur (Figure 1A). Both consistency checks of §2.2 held: the summed per-event DLPs agreed with the independently recorded accumulated DLP to within 1% in 14 of 14 comparable files (Figure 1B), and the number of events parsed equalled the number declared in 14 of 14 files.

3.2 Recorded-dose availability in a public archive

CTDIvol was absent from the image header in 206 of the 400 screened series (51.5%), but the pooled figure conceals the dominant effect (Figure 2A): 100.0% for GE, 48.0% for Siemens, 34.0% for Philips and 24.0% for Canon/Toshiba. Availability of a recorded CTDIvol was strongly associated with manufacturer ($\chi^2(3) = 137.2$, nominal $p < 0.001$, Cramér's V = 0.59); because series from one collection may not be independent, that p-value should be read cautiously and the effect size emphasized. No series carried a usable DLP record in the dose comment.

Of the two conditions for reconstruction (§2.5), the acquisition description was complete enough in 52.5% of the 400 series and the scanner tabulated in 28.8%. **Of the 206 series that lacked a recorded CTDIvol, 42 (20.4%) retained enough metadata to reconstruct one**, a conditional rate far more manufacturer-dependent than the marginal one (Table 1). By either route 236 of 400 series (59.0%) were dose-recoverable, leaving 164 (41.0%) no method here can recover.

Table 1. Recorded-dose availability and reconstructability by manufacturer, in 400 public CT series (100 each).

Manufacturer	CTDIvol absent	Of those, reconstructable	Conditional rate
GE	100 / 100 (100.0%)	37	**37.0%**
Siemens	48 / 100 (48.0%)	0	**0.0%**
Philips	34 / 100 (34.0%)	0	**0.0%**
Canon/Toshiba	24 / 100 (24.0%)	5	**20.8%**
All	**206 / 400 (51.5%)**	**42**	**20.4%**

Tube current, slice thickness and slice location were present in nearly every series; total collimation and pitch were not, and those two limit feasibility (Figure 2B). Scanner coverage is the larger constraint: the table resolved 115 model names, found 0 ambiguous and did not cover 285, and a further 122 series were described completely enough to reconstruct but sat on a scanner the table lacks.

3.3 Accuracy against the archive's own recorded values

Pooled over all 46 evaluated series the median relative error was −8.7%, with a mean absolute error of 21.7%. The 46 comprise 45 distinct (recorded, reconstructed) pairs and divide **Canon/Toshiba 27,**

Siemens 12, Philips 7, GE 0 — GE contributing nothing not for want of coverage, being the best-covered manufacturer, but because no GE series in the sample carries a recorded CTDIvol at all (100.0% absence); §3.4 addresses GE separately.

In the pre-specified primary stratum, whose headers record the total collimation: n = 40, median relative error −5.7% (95% percentile-bootstrap CI −12.4% to +7.8%), mean absolute error 22.0%, and 60.0% of cases within ±20% (Figure 3). The interval contains zero, so the spread rather than the center is the result; of the 40, 39 are distinct pairs from 21 collections, so it may underestimate uncertainty.

The error is two-sided, not conservative. Reconstruction over-estimated the recorded CTDIvol in 18 of the 46 cases and under-estimated it in 28, spanning −59.4% to +87.4%; the largest over-estimate reconstructed 16.68 mGy where the scanner recorded 8.90 mGy (Figure 3A). A reconstructed CTDIvol therefore cannot be treated as a lower bound, nor can a DLP or effective dose derived from one.

With no bowtie recorded, 45 of the 46 assumed the conventional filter and seven also took the nearest tabulated collimation: **no case matched a tabulated technique exactly**. Per-manufacturer and per-stratum detail, the size-correction and effective-dose results, the uncovered models and the data-quality defects found in the published tables are in Supplementary Material (§S2).

3.4 Accuracy on GE, against the scanners' own dose reports

The three GE objects hold 35 irradiation events, 15 of which record a CTDIvol. By the criteria of §2.7, 4 of those 15 were excluded because the table does not move, 4 for a geometry that contradicts their recorded type, and 2 because their scanner is a LightSpeed RT16, the radiotherapy configuration (§S2.5); 5 events from 2 scanner models were evaluated (Table 2).

The scanner's own exposure time refutes the collimation four of the five events record. Both objects state 2.5 or 5.0 mm on 64-slice scanners, where the measured time gives 18.6 to 36.4 mm — a difference of 86% to 87%; for the fifth it gives 39.8 mm against a recorded 40.0 mm, agreeing to 0.6%. On the four, the recorded field is not a beam width — most plausibly a detector row or a nominal reconstructed thickness — a defect in what the scanner writes, not in the table.

Where the recorded width is refuted the span runs over every tabulated row; where it stands, over the filter alone (Table 2). The four refuted events span −18.2% to +22.4%, or −12.9% to +3.8%

under the narrower reading, and the corroborated one −0.0% to +12.7%, its recorded 40 mm row reproducing the recorded value exactly (22.26 mGy against 22.26 mGy, −0.0%) where giving that row up too would reach +35.3%. **Some admissible combination reproduces all five recorded values to within 0.5%** — the pre-stated condition for counting the arithmetic reproduced, against a 1% tolerance from the rounding of the two numbers compared, met under either reading (§S2.5). Inverting the formula gives an implied coefficient of 8.48 to 8.74 mGy per 100 mAs across the four 32 cm events, narrower than any of these spans, placing the residual in the table lookup rather than the formula (§S2.5).

Table 2. Reconstruction against the CTDIvol recorded by GE scanners in their own dose reports, for the five events meeting the criteria of §2.7, at 120 kVp. The measured collimation is $Lt_{rot}/(Tp)$. Where it contradicts the recorded width, that width is refused and the span covers every tabulated collimation and filter; where it agrees, the recorded width stands and only the filter varies.

Scanner	Phantom	Recorded (mGy)	Collimation: stated / measured (mm)	Verdict	Span over assumptions
Optima CT660	32 cm	3.23	5.0 / 34.5	refuted	−12.9% to +21.3%
Optima CT660	32 cm	5.30	5.0 / 36.4	refuted	−12.1% to +22.4%
LightSpeed VCT	32 cm	8.74	2.5 / 19.0	refuted	−17.9% to +16.4%
LightSpeed VCT	32 cm	4.93	2.5 / 18.6	refuted	−18.2% to +16.1%
LightSpeed VCT	16 cm	22.26	**40.0 / 39.8**	**stands**	**−0.0% to +12.7%**

4. Discussion

Availability differed among manufacturers by up to 76 percentage points. Model and export configuration may contribute, but the association is strong enough that any pooled "public CT data lacks CTDIvol" misleads in both directions.

The coefficient constraint is worth separating from the metadata one, because the two have different remedies: more series were complete but untabulated than the reconstruction could run on (§3.2). Extending an open coefficient database is therefore the single change that would most widen

retrospective dose recovery, and it is a metrology task, not a software one. I did not close it by matching an uncovered model onto a nearby covered one: a plausible substitution converts a refusal into a silent error, which cost two eligible GE events (§3.4).

The GE arm adds a point the archive cannot make: **a recorded acquisition parameter can be wrong, and the same report can show that it is**. Width is the parameter the dose model is most sensitive to, and four of five events state one their own exposure time refutes, so a reconstruction reading that field without a cross-check inherits the error silently. Where the report is self-consistent the table reproduced the scanner's own index exactly; where it is not, the honest output is a span — failing loudly rather than quietly. And a cross-check that refutes a parameter cannot always replace it: that the recorded width is wrong by 86% needs no precision, but naming the row the scan used would need a few per cent the report does not support, so the width becomes undetermined and the range widens rather than narrows.

Where an RDSR exists, this parser reads the same content items OpenREM harvests [4]; what is added is the case where none does. The contribution is not another CTDIvol calculation but a decision framework: when retrospective recovery is defensible, which assumptions govern it, and when none hold. A reconstructed index is usable because its provenance and uncertainty travel with it, and unusable once quoted as recorded. An organ-weighting layer and a patient-specific extension are the subject of separate work.

Limitations. Every reconstructed value rests on an assumed bowtie filter (§2.7). Scanners carry filters beyond the three the table holds — for pediatric patients, cardiac acquisitions and differently sized bodies — none recorded in DICOM, so the assumption cannot be checked against the data. Its weight is bounded by the table itself: across the 605 comparisons in which more than one filter was tabulated for the same model, voltage, collimation and phantom, substituting a different filter moves nCTDIw by a median of 12.2%, with 82% within 20% and an extreme of +110%. A scanner whose clinical filter departs systematically from the conventional one would be biased throughout.

The GE arm is small and indirect: five events on two models, not independent patients, validating the dose model and not recovery from a de-identified header. Only one geometry is corroborated, and every result is a span rather than a point. The GE conditional rate in Table 1 remains unvalidated against a header-derived reference, because none exists. Reconstructed DLP omits over-ranging and is not a

bound in either direction (§2.3). Effective doses here carry ICRP 60 weighting through the Report 96 coefficients [8] and would differ under ICRP 103 [9,10]. The RDSR arm rests on 14 objects from one collection with no external ground truth, and the table did not cover 285 of 400 sampled series, so the accuracy arm is neither representative nor a validation across these scanner ranges. Most fundamentally, **neither accuracy arm has an independent reference**: the recorded CTDIvol is a console figure and the coefficient table is a survey of console figures from the same factory calibrations [7], so the agreement tests the arithmetic from metadata to dose index and not the dosimetry beneath it. Where a technique coincides with a tabulated row, as on the corroborated GE event, exact agreement is close to arithmetic.

5. Conclusions

A CT dose index can be recovered from acquisition metadata when no recorded value survives, but only as far as that metadata and the open coefficient table allow, and a recovered value is not interchangeable with a recorded one. Formula-, input-, assumption- and source-level provenance keeps that distinction explicit and the uncertainty attributable. A recorded index was missing from roughly half the series screened, and four in ten admitted no estimate at all. The error is two-sided, so a recovered index is not a conservative bound. A recorded acquisition parameter can itself be wrong and the same report may show it, so a reconstruction should cross-check what it reads — and check the alternatives too. Availability should be reported by manufacturer.

Declarations

Conflicts of interest. The author is the representative of LISIT Co., Ltd. (Tokyo, Japan) and Chief Executive Officer of TexelCraft OÜ, and this work was conducted under the supervision of LISIT Co., Ltd. The software is released under the MIT License and the author has a commercial interest in downstream products that may use it. No patient or customer data were used.

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Ethics approval and consent to participate. Not applicable. This study used only publicly available, de-identified imaging data — TCIA collections released under Creative Commons licenses [13] — and

third-party sample files, so no review board approval or informed consent was required. No imaging is redistributed: the series analyzed stay in their public collections and the RDSR samples with their GPL-3.0 distributor, and the coefficient table under its own CC BY 4.0 license [7]. No private or internally anonymized images were used.

Data and code availability. The software is available at https://github.com/Institute-of-One/ctdose-core under the MIT License. The version used in this study (version 0.1.1) is archived at Zenodo (https://doi.org/10.5281/zenodo.21636719). All numerical results reported here, including the rows they summarize, ship under `results/` with the scripts that generate them and the figure code.

Use of artificial intelligence. A generative AI assistant (Claude, Anthropic) supported software development and drafted portions of this manuscript. All results were produced by executable code in the released repository and re-run and verified by the author, who takes full responsibility. The AI system is not an author and does not meet authorship criteria.

Author contributions. S.Y.: conceptualization, methodology, software, validation, formal analysis, investigation, data curation, writing — original draft, writing — review and editing, visualization.

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Figure legends and figures

Figure 1. *Vendor-neutral parsing of recorded dose from real CT Radiation Dose Structured Reports.* **(A)** Per-event CTDIvol from 14 objects spanning four manufacturers; marker shape is the stated reference phantom. **(B)** Sum of the per-event DLPs against the accumulated DLP recorded independently in the same object.

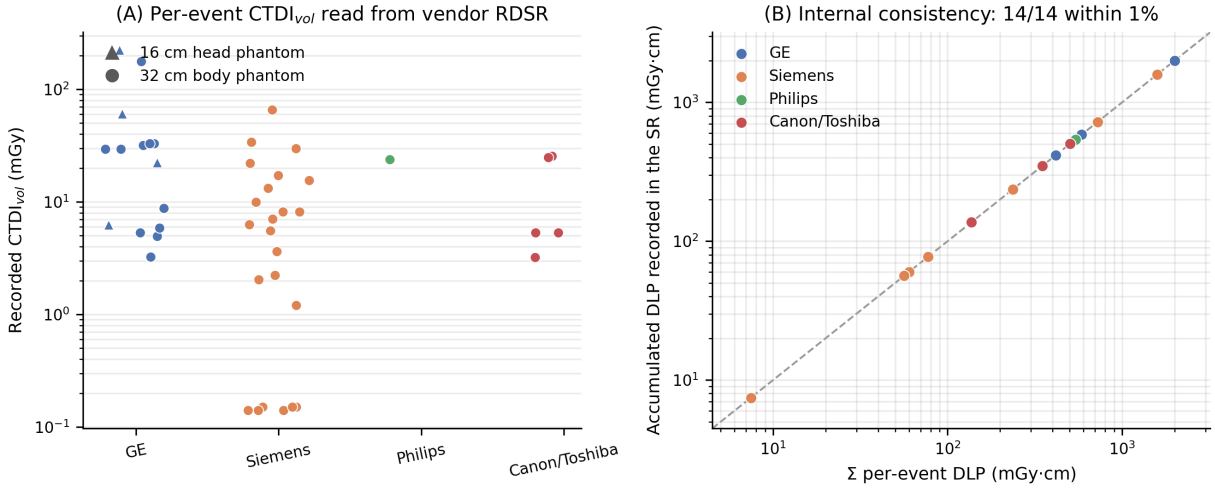


Figure 2. *Recorded-dose availability in 400 public CT series, 100 per manufacturer, across 92 TCIA collections.* **(A)** Percentage with CTDIvol absent from the header, with metadata complete enough to reconstruct it, and recoverable either way. **(B)** Percentage carrying each acquisition attribute.

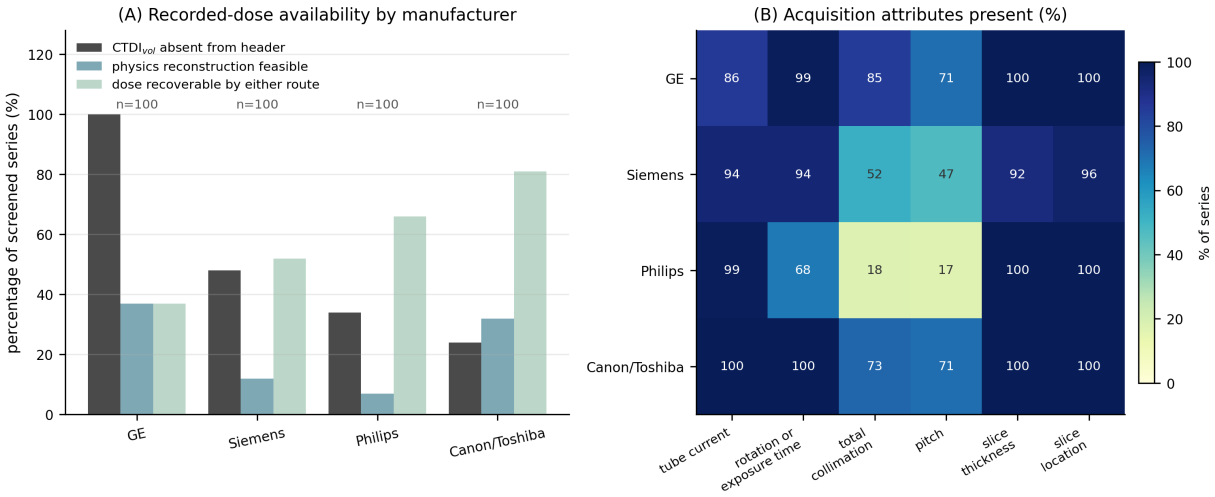


Figure 3. *Accuracy of the reconstruction against the scanner's recorded CTDIvol (n = 46).* **(A)** Reconstructed against recorded, identity line dashed, factors of two dotted; circles are the primary stratum. Points above the identity line are the 18 over-estimates. **(B)** Relative-error distribution by stratum, median marked.

